Sensor Applications

HFR-video-based 3D Software Sensor for Bridge Displacement Monitoring

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Abstract—In this paper, we propose a novel vision-based 3D software sensor using high-frame-rate (HFR) videos for bridge measurement that can realize non-contact full-view 3D dynamic analysis, including displacement distribution and frequency response. Model-based 3D-to-2D projection is applied for addressing weak-texture bridge images to effciently match corresponding points in stereo HFR 4K videos. Then relative 3D displacement is precisely calculated based on a pair of 2D displacement vectors, which are estimated on each camera by digital image correlation with sub-pixel accuracy. 3D displacement signals are analyzed by short-time Fourier transform for vibration analysis to calculate frequency responses of all measured points, and 3D mode shapes of peak frequencies that refer to intrinsic structure characteristics are estimated in frequency domain. To verify its effectiveness, a prestressed concrete girder bridge of 23 m length in Japan was measured and the experimental results show that our software sensor can achieve multi-point 3D displacement measurement with an accuracy of 0.01 mm and 3D vibration analysis in the frequency range of 0–50 Hz.

Index Terms—Structure Health Monitoring, High-speed Vision, 3D Model, Digital Image Correlation, Vibration Mode Analysis.

I. INTRODUCTION

Most bridges are built around rivers, mountain terrain, and highway roads, where poor environmental conditions such as earthquakes, long-term sunshine, wind gusts, and moving traffic loads will break the bridge structure and cause serious safety accidents together with huge economic loss. Bridge displacement measurement is important for estimating the dynamic response of bridges to monitor their structural health condition and predict appropriate maintenance time. However, bridges are typically large in size and volume, making it challenging to achieve full-view 3D measurements of their dynamic response. In recent years, many studies have been conducted to realize bridge displacement measurement; they can be mainly divided into contact sensor-based methods and non-contact sensor-based ones.

Contact sensor-based methods need sensors installed on the surfaces of bridges to sense their deformation directly. The widely-used sensor is the linear variable differential transformer [1][2]; it is installed between the target and a reference point to measure the deflection by converting its core deformation into electric voltage signals. Strain gauges [3][4] can be used for bridge measurements by converting their resistance value, which changed with deformation, into a digital voltage value. These electrical signals are transmitted to computers by cables, which seriously limit the distance between the computers and the bridge. To solve this problem, wireless sensors [5][6] have been applied by utilizing micro-electro-mechanical and wireless sensing technologies to realize low-cost and flexible measurement. These sensors require direct installation on the bridge, which is difficult because of difficult-to-reach locations, and they only support single-point measurement, which is insufficient for 3D global estimation, such as mode shape computations.

Non-contact sensor-based methods can avoid contact installation requirements to provide a friendlier operation. Global positioning system (GPS) [7][8] is used for displacement measurement with triangulation by setting a receiver on the bridge to calculate the distance from different satellites. However, its estimation accuracy is relatively low at a level of 1–2 cm [9]. Laser Doppler vibrometers [10][11] can achieve sub-micron-level measurement accuracy with a wide frequency range up to 20 kHz, whereas it is very expensive and the laser intensity will be dangerous for people working in real space. In recent years, vision-based methods [12][13] have been used for full-view displacement measurement, and 3D digital image correlation (DIC) methods [14][15], have been applied for 3D displacement estimation by calculating the difference between images before and after deformation. Most 3D DIC methods work in friendly laboratory environments, where rich textures are attached to the target such that small region of interest (ROI) sizes can work. The accuracy in these 3D DIC methods relies heavily on the appearance information for stereo matching to correspond points between different views, whereas the most bridges have little texture for 3D DIC. Moreover, most studies of vision-based bridge monitoring have used cameras operating at 25 or 30 fps; they are insufficient for small-size bridges of iron- or concrete girders to analyze displacements with resonant frequencies at tens of Hertz.

In this paper, we propose a vision-based 3D software sensor that can enlarge the measurement frequency range using high-frame-rate (HFR) cameras and improve the robustness against outdoor textureless cases by introducing model-based stereo matching. For dynamic displacement analysis at thousands of 3D software sensors, DICs and short-time Fourier transforms (STFTs) have been accelerated with parallel implementation, and its effectiveness was demonstrated with 3D deformation images of tens-of-meters length bridge when a heavy truck passed on it.

II. HFR-VIDEO-BASED 3D SOFTWARE SENSOR

A. Concept

Current 3D DIC methods have two limitations in bridge displacement measurement; (1) most of bridges in outdoor scenes have not always sufficient patterns on their surfaces, whereas 3D DIC methods assume appearance-based stereo matching that require a rich texture on the object surface to find corresponding points between different cameras; (2) 3D displacement estimation at thousands of points using 3D DIC methods is too time-consuming to process high-resolution

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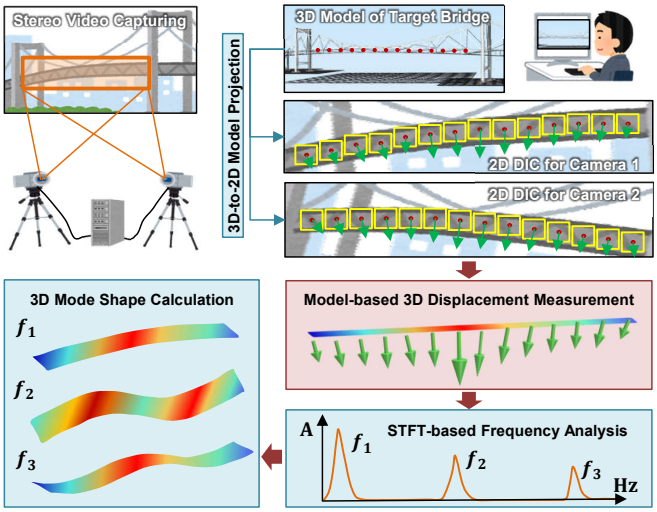

Fig. 1. HFR-video-based 3D software sensor using a 3D model.

HFR images in long-time and real-time bridge monitoring, because of their computational complexities in pixel-wise stereo matching.

To improve the robustness against texture-less cases and accelerate 3D displacement estimation in bridge monitoring, we propose a novel concept of HFR-video-based 3D software sensor for semi-rigid objects with small displacements and vibrations. Assuming that the object to be measured is not largely deformed at a fixed location and its 3D shape is priorly given as a 3D model, thousands of measurement points are virtually assigned on the 3D model as software sensor locations, and relative 3D displacements from the 3D model are estimated at all points without any pixel-by-pixel stereo correspondence process between two cameras. The procedure of our 3D software sensor is divided into the following three steps, as illustrated in Fig. 1.

(1) Stereo Video Capturing with 3D-to-2D Model Projection

Stereo HFR videos with different views are recorded by high-speed cameras. The 3D shape is priorly measured as its 3D model and multiple 3D points selected on the model are projected on the two 2D camera views with pre-calibrated parameters.

(2) Model-based 3D Displacement Measurement with DICs

2D DICs at the corresponding points on the two cameras are simultaneously employed with GPU acceleration to estimate their 2D displacements. The 3D displacements at all points are relatively computed with triangulation using these 2D displacements.

(3) 3D Vibration Analysis with Point-wise STFT Computation

Frequency responses of 3D displacements are calculated with STFT computation at all points. 3D mode shapes at their peak frequencies are estimated as intrinsic structure characteristics in frequency domain.

B. Implemented Algorithms

We assume that a 3D model is given for the object to be measured as a set of 3D points, $\mathcal{P} = \{(X_i, Y_i, Z_i); i = 0, \ldots, N-1\}$, and M pairs of 2D points on left and right cameras, $\{(\hat{x}_m^L, \hat{y}_m^L), (\hat{x}_m^R, \hat{y}_m^R)\}$, are priorly corresponded to the 3D points on the model, $(\hat{X}_m, \hat{Y}_m, \hat{Y}_m)(\in \mathcal{P})$ $(m = 0, \cdots, M-1)$, which are manually selected for camera calibration. Corresponding to the number of the 3D points belonging to $\mathcal{P}$, the following algorithms are parallel-implemented to process the input images of the left and right cameras, $I_L(x, y, t)$ and $I_R(x, y, t)$, which are captured at intervals of τ.

1) 3D-to-2D Model Projection: Calibrating with the pre-selected M pairs of corresponding points, all the 3D points (X_i, Y_i, Z_i) $(\in \mathcal{P})$ on the 3D model are uniquely projected at the following 2D points on the image planes of the left and right cameras,

$$(x_i^L, y_i^L) = f^L(X_i, Y_i, Z_i), \quad (x_i^R, y_i^R) = f^R(X_i, Y_i, Z_i), \tag{1}$$

where $f^L(\cdot)$ and $f^R(\cdot)$ indicate 3D-to-2D projection functions, which are determined by the calibrated camera parameters.

2) Sub-pixel Displacement Estimation with DICs: 2D displacements are independently estimated on the left and right cameras by parallel-implementing 2D DICs [16] for ROI (region of interest) images around all the 3D-to-2D projected points as follows:

$$(u_i^L(t), v_i^L(t)) = \mathrm{DIC}(I^L(x, y, t), I_0^L(x, y); (x_i^L, y_i^L)), \tag{2}$$
$$(u_i^R(t), v_i^R(t)) = \mathrm{DIC}(I^R(x, y, t), I_0^R(x, y); (x_i^R, y_i^R)), \tag{3}$$

where $\mathrm{DIC}(\cdot)$ indicates a 2D DIC function that can estimate relative displacement between two images with sub-pixel accuracy. $I_0^L(x, y)$ and $I_0^R(x, y)$ are reference images in DIC computation.

3) 3D Displacement Measurement with Triangulation: 3D displacements at all points are calculated with triangulation by adding the 2D displacements estimated on the left and right cameras with the coordinate values of the 3D-to-2D projected points as follows:

$$(U_i(t), V_i(t), W_i(t)) = F((x_i^L + u_i^L(t), y_i^L + v_i^L(t)), (x_i^R + u_i^R(t), y_i^R + v_i^R(t))), \tag{4}$$

where $F(\cdot)$ is a triangulation function for 3D point estimation from two camera views; it is determined by the calibrated transformation matrix between the 3D model and the 2D image. For each camera, 4×4 transformation matrix is estimated by manually selected 10 3D points in the bridge model and their corresponding 2D points in the camera image. Relative 3D displacements at all points, $(\Delta U_i(t), \Delta V_i(t), \Delta W_i(t))$ $(i = 0, \cdots, N-1)$, are computed by differentiating the 3D model shape from estimated displacements.

$$(\Delta U_i(t), \Delta V_i(t), \Delta W_i(t)) = (U_i(t) - X_i, V_i(t) - Y_i, W_i(t) - Z_i). \tag{5}$$

4) 3D Vibration Analysis with STFT Computation: To verify 3D vibration distribution in frequency domain, frequency responses are estimated at all the 3D points by computing the STFTs of these 3D displacements in parallel as follows;

$$\boldsymbol{F}_i(t) = \mathrm{STFT}(A_i(t), A_i(t+\tau), \ldots, A_i(t+(K-1)\tau)), \tag{6}$$
$$\bar{\boldsymbol{F}}(t) = \frac{1}{N}\sum_{i=0}^{N-1} \boldsymbol{F}_i(t), \tag{7}$$

where $A_i(t)$ indicates the X-, Y- or Z-component of the relative 3D displacement vector estimated at (X_i, Y_i, Z_i) $(i = 0, \cdots, N-1)$. $\boldsymbol{F}_i(t)$ is the frequency response at (X_i, Y_i, Z_i), which is obtained by executing STFTs with K frames. $\bar{\boldsymbol{F}}(t)$, the averaged frequency response of all N points, is used to detect peak frequencies for resonant vibrations. Relating to the resonant vibrations, the 3D mode shapes at the peak frequencies f_p $(p = 0, \ldots, P-1)$ can be computed in time domain by executing inverse STFTs for the frequency components at f_p at all points.

C. Configuration and Performance

These algorithms have been parallel-implemented on a GPU-based high-speed vision platform. It consists of two high-speed camera heads (EoSens 21CXP2, Mikrotron, Unterschleissheim, Germany) with a CXP-12 frame grabber, and a personal computer with a CPU

TABLE 1. Execution time (unit: ms)

3D Points	100	1000	2000	5000
2D DICs	5.05	41.34	80.55	195.12
3D Triangulation	0.08	1.54	2.12	4.38
3D STFTs	0.25	0.75	0.92	1.61
Total Time	5.13	42.88	82.67	199.50

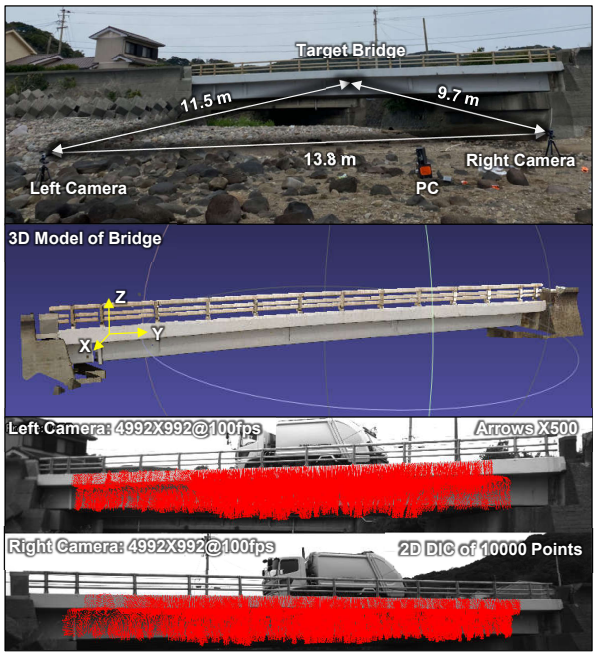

Fig. 2. Experiment setting when monitoring Tsubami bridge.

board (Intel Core i9-14900KF@3.20GHz) (ASRock Z790 Pro RS, Taipei, Taiwan) for HFR video recording and processing, which is accelerated by a GPU board (RTX 6000 Ada Generation, NVIDIA, Santa Clara, CA, US).

Table 1 shows the execution time when 4992×994 images captured at 100 fps, as used below in the bridge experiment, are processed for 3D displacement and vibration analysis at 100, 1000, 2000, and 5000 points; 2D DICs and 3D STFTs are parallelized on the GPU, whereas 3D triangulation is executed on the CPU. 2D displacements are estimated by 2D DICs with 256×256 neighborhood regions, and frequency responses are obtained by STFT computations with 128 frames. This shows that our software sensor can operate in real time when only hundreds of 3D points are measured, and it can operate within 200 ms when the number of 3D points reached 5000, which shows good efficiency for off-line processing.

III. EXPERIMENT

To verify the performance of our 3D software sensor, we recorded stereo videos of 4992×994 pixels at 100 fps when a heavy truck passed on Tsubami Bridge, which is a 2-lane prestressed concrete (PC) girder bridge of 23 m length in Nagasaki Prefecture, Japan. The stereo HFR 4K videos were analyzed offline to visualize its full-view 3D deformation and vibration distribution estimated at 10000 points, which were set on the 3D model of the bridge.

Fig. 2 shows the experimental setting when monitoring Tsubami bridge. As shown in the upper photo, two high-speed cameras with a

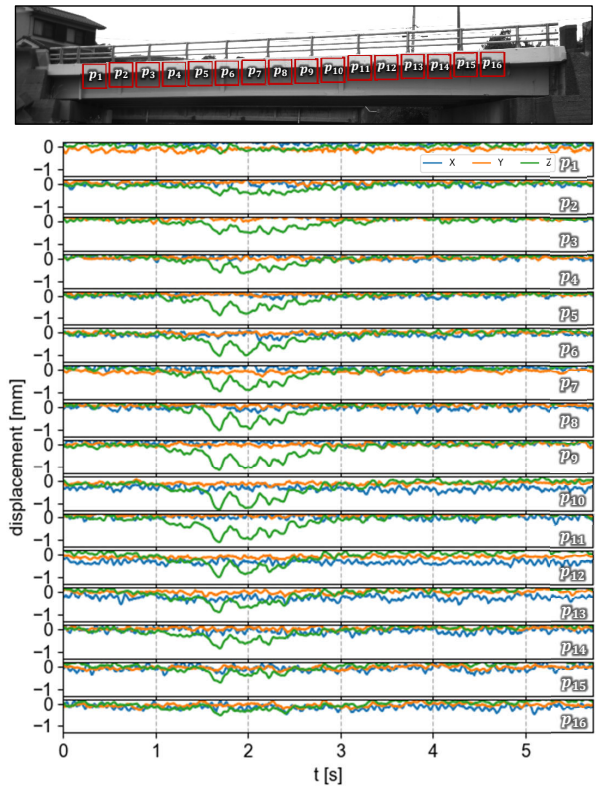

Fig. 3. Displacement signals of 16 points when a truck passed.

distance of 13.8 m apart were set approximately 10 m from the center of the bridge. The laser scanner (Focus S150, FARO Technologies, Lake Mary, FL, US) 3D-scanned the bridge with 2668369 points as shown in the middle photo, and the 3D model was resampled for 3D software sensors at 10000 points. 8-minute synchronized stereo videos of 4992×992 pixels were recorded at 100 fps, including the times when several large trucks passed on it. Both the 4992×994 images of the left and right cameras wholly covered the side-view of the 23 m length bridge; one pixel corresponded to 5 mm or less when observing its center. Generally, measurement of longer bridges requires cameras with larger resolutions, and camera resolutions should be appropriately selected for sub-millimeter-level displacements to ensure one pixel corresponds to 5 mm or less, according to the length of bridges to be measured. The upper photo shows the input images of the left and right cameras when a truck passed on the left lane of the bridge, including 500 times magnified arrows of 2D displacements estimated by DIC with 256×256 neighborhood regions at 10000 points.

Fig. 3 shows the 3D displacements at the 16 selected points ($\boldsymbol{p}_1$–$\boldsymbol{p}_{16}$) for 5.5 s. Blue, yellow, and green lines represent the displacement in the x, y, and z directions, respectively. A heavy truck passed on the center of the bridge around t = 1.7 s. Corresponding to the passage of the truck, the displacement of 1 mm amplitude was observed with a certain vibration in the Z direction. The vertical displacement increased from both ends towards the middle, reaching its maximum value of 1.1 mm at $\boldsymbol{p}_{10}$, which was located around the center. Estimating 3D displacement vectors at 10000 points on the 3D model, Fig. 4 shows the deforming 3D shapes of the bridge for t = 0.0–5.0 s; the displacement ranges from 0.0 to 1.1 mm in the Z direction. The relative displacements from the 3D model was 1000 times magnified, and they were enhanced by a color gradient from blue to red. The truck began crossing the bridge from the right side at

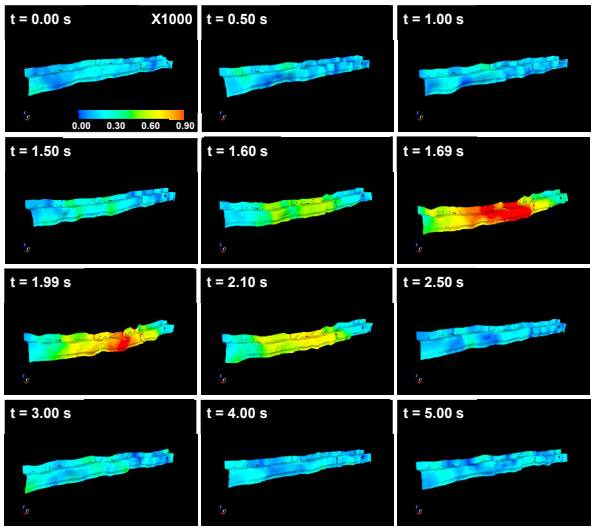

Fig. 4. 3D deformation images when a truck passed.

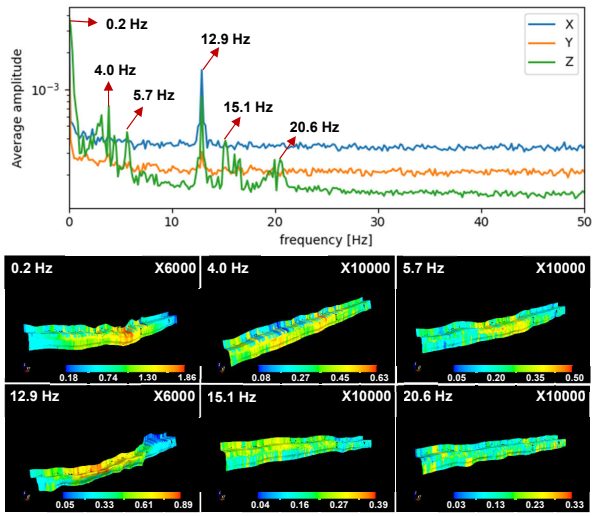

Fig. 5. Frequency response and mode shapes at peak frequencies.

t = 1.50 s. As the bridge bore the full weight of the truck, the bridge reached its maximum displacement by t = 1.69 s. Subsequently, the bridge's displacement decreased gradually, returning to zero, as the truck exited from the left side.

Fig. 5 shows the averaged frequency response of all 10000 points, and the 3D mode shapes at several peak frequencies. 0.2 Hz keeps the maximum energy that infers the main deformation of the bridge, where the middle region sank in the Z direction. 12.9 Hz may correspond to the resonant vibration excited by continuous seaside wind, which deforms the bridge in the X- and Z- directions. The left 4.0, 5.7, 15.1 and 20.6 Hz should be the frequency components that are caused by the moving load on the bridge; they are mainly vibrated rigidly with smaller local deformations.

These results demonstrated that our 3D software sensor can achieve non-contact and full-view displacement monitoring of real bridges and provide dynamic analysis results with high spatiotemporal resolution. Precise displacements of thousands of points can intuitively illustrate bridge deformation and support high-resolution mode analysis, which is crucial for structural analysis of bridges.

IV. CONCLUSION

In this paper, we proposed a novel concept of HFR-video-based 3D software sensor for non-contact full-view bridge monitoring that can improve the robustness against texture-less surfaces and accelerate dynamic 3D displacement analysis for a semi-rigid object. Thousands of measurement points can be virtually assigned on a 3D model as software sensor locations, and it enables to precisely estimate the 3D deforming shape without any pixel-by-pixel stereo matching. We applied our 3D software sensor to a PC girder bridge in Japan, and the experimental results show the 3D displacement measurement with an accuracy of 0.01 mm and frequency response estimation in the range of 0–50 Hz. This paper focused only on offline video analysis, and there is a strong need for real-time bridge monitoring. In the future, we will accelerate the current video-based software sensor with fewer points and conducting a series of real-time experiments, and extend our system using more than two cameras for more accurate and larger-range measurement. Road traffic information with AI detection is also planned to be combined with current dynamic analysis for more intuitive feedback information.

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